Critical Thinking

Reflective Journal

Self-Reflection on Key Learnings and Professional Application

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Introduction

This journal captures reflections on critical thinking concepts explored during the course. Each entry examines a specific dimension, including cognitive biases, thinking systems, the role of emotions, and structured problem definition. The goal is to move beyond passive understanding toward active, deliberate application of these principles in day-to-day decision-making.

1. Identifying Biases from Past Experience

Question: *Identify the biases that you are now able to relate to in your past experience. List the top three biases.*

Looking back at my experience as a full-stack developer, three biases clearly affected my decisions:

- **Confirmation Bias:** When choosing between Angular libraries or .NET packages, I often favoured options I had used before and ignored alternatives suggested by teammates.
- **Anchoring Bias:** During sprint planning, the first time estimate for a task would stick in everyone's mind. Even when we realised the feature was more complex, the original number kept pulling decisions back.
- **Availability Bias:** After dealing with a painful production bug like a memory leak in a .NET API, I would over-focus on preventing that same issue while overlooking common risks like SQL injection.

Recognising these biases now helps me make more balanced decisions.

2. Fast (Non-Conscious) vs. Slow (Deliberate) Thinking

Question: *Did you catch yourself when you went through Fast (Non-Conscious) Thinking and Slow (Deliberate) Thinking? Share your experience.*

Yes. After learning about fast and slow thinking, I began noticing both modes in my daily work.

When a critical bug appeared in our Angular application, my fast thinking told me the issue was in a recently changed component. I was about to start debugging there without checking anything else. But I paused and switched to slow thinking. I opened the browser developer tools, traced the network calls, checked the C# API logs, and found the real problem was a null reference in a .NET service method, completely unrelated to the Angular code.

That pause saved me hours of debugging in the wrong place. Fast thinking helps us react quickly, but for complex bugs, we need to slow down and follow the data.

3. The Role of Emotions in Critical Thinking

Question: *Do you think eliminating emotions completely would be the right way to think critically? Share your experiences.*

No. Removing emotions completely is neither realistic nor helpful. Emotions act as early warning signals that point us toward things needing attention.

During a code review for a C# API refactoring, I felt uneasy about the approach even though the code compiled and all tests passed. Instead of ignoring that feeling, I raised my concern. On closer review, we found the new design would break backward compatibility for several Angular front-end consumers depending on the old response structure.

This taught me that the right approach is not to ignore emotions but to use them as a starting point. Feel the signal, then verify it with evidence before making a decision. Emotional awareness and critical thinking work best together.

4. Redefining a Past Problem Statement

***Question:** Pick a problem or bug that you worked on a few months back. Now with this new awareness, how would you define the problem statement better if the problem was given to you today? Which method helped you to define the problem statement better?*

A few months ago, our team dealt with a frustrating issue: “The Angular app is slow.” This vague statement sent us in multiple directions and we wasted days without progress.

Using the **5 Whys method** from this course, I would now approach it differently. Why is the app slow? Because certain pages take over five seconds to load. Why? Because multiple API calls fire at once during component initialisation. Why? Because there is no caching in the Angular service layer. Why? Because the services were built for a smaller dataset that has since grown.

The refined problem statement would be: “Simultaneous uncached API calls during page load cause response times above five seconds on data-heavy pages.” This is specific, measurable, and actionable. The 5 Whys method was key to reaching this clarity.

Conclusion

This exercise reinforced that critical thinking is a set of everyday habits: spotting biases, pausing before reacting, using emotions as signals rather than drivers, and defining problems clearly before solving them. As a developer, these habits directly improve how I debug, review code, and make design decisions. Going forward, I plan to consciously apply these principles in my daily work.